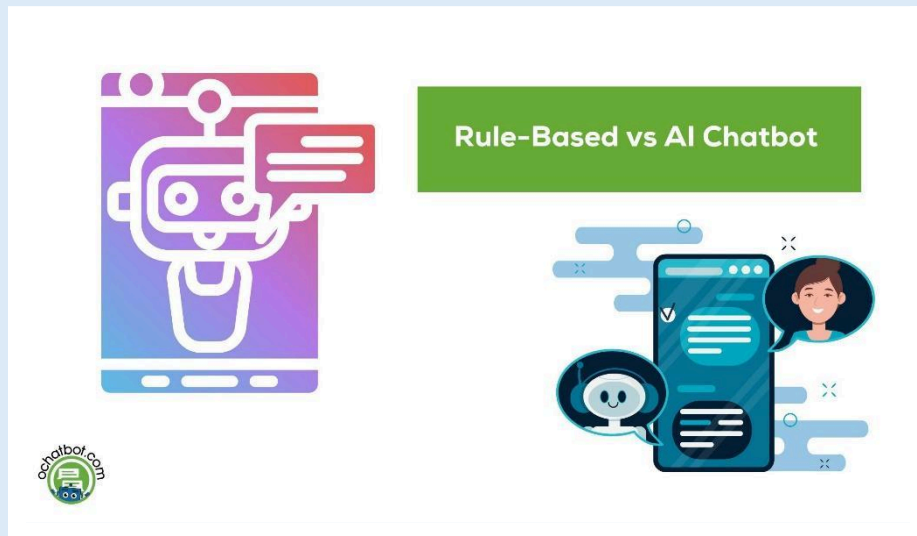


Project Name: Rule-Based AI Chatbot



Internship: Artificial Intelligence. Industrial training

Organization: Decode labs

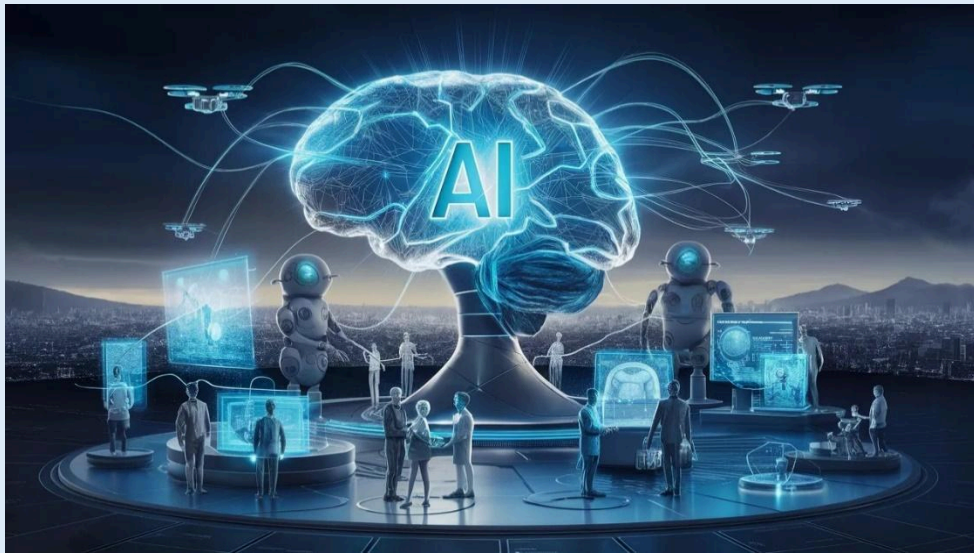
Student Name: Prabhjyoti

Roll Number: 2452127

Submission Date:

Introduction

Artificial Intelligence (AI) enables machines to perform tasks that normally require human intelligence. One of the simplest applications of AI is a rule-based chatbot. A rule-based chatbot responds to user inputs using predefined rules and decision-making logic. This project demonstrates the use of conditional statements and loops to simulate a basic conversation between a user and a computer system.



Objective

The main objective of this project is to develop a simple Rule-Based AI Chatbot using Python. The chatbot interacts with users by recognizing predefined inputs and providing appropriate responses.

Objective 1: Create a Simple Chatbot

- Design a chatbot that can communicate with users through text messages.
- Allow users to enter queries and receive responses.

Objective 2: Understand Decision-Making Logic

- Use **if-else statements** to check user inputs.
- Make the chatbot choose different responses based on different conditions.

Objective 3: Implement Continuous Interaction

- Use a **while loop** to keep the chatbot running continuously.
- Allow multiple conversations without restarting the program.

Objective 4: Handle Greetings and Basic Commands

- Recognize common greetings such as "Hello", "Hi", and "Hey".
- Respond with appropriate greeting messages.

Objective 5: Process Exit Commands

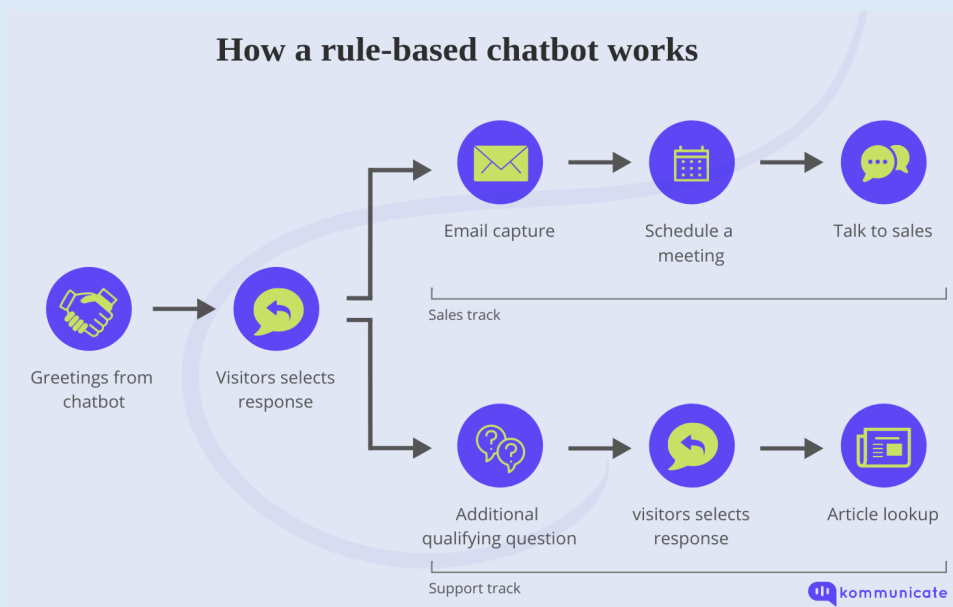
- Detect commands like "Bye", "Exit", or "Quit".
- End the conversation gracefully when the user wants to leave.

Objective 6: Build a Foundation for AI Concepts

- Learn the basics of chatbot development.
- Understand how AI systems can simulate simple human conversations using predefined rules.

Overall Objective:

To gain practical knowledge of Python programming, control flow, loops, and conditional statements by developing a basic AI chatbot that can interact with users and respond according to predefined rules.



Problem Statement

In today's digital world, users expect quick and interactive communication with computer systems. However, computers cannot understand human language naturally without proper instructions. Therefore, there is a need to develop a chatbot that can respond to user inputs using predefined rules.

The problem is to design and implement a **Rule-Based AI Chatbot** that can:

- Accept user input.
- Identify predefined keywords or commands.
- Generate appropriate responses using **if-else conditions**.
- Continue the conversation in a loop until the user chooses to exit.

This project aims to demonstrate how basic Artificial Intelligence concepts and decision-making logic can be used to simulate simple human-computer interaction.

Software Requirements

The following software tools are required to develop and run the Rule-Based AI Chatbot:

1. Python 3.x

- Python is the programming language used to develop the chatbot.
- It provides simple syntax and built-in functions for implementing chatbot logic.

2. Code Editor / IDE

- A code editor such as **Visual Studio Code (VS Code)**, **PyCharm**, or **Jupyter Notebook** is used to write, edit, and execute the Python program.

3. Operating System

- The chatbot can run on **Windows, Linux, or macOS** systems that support Python installation.

4. Command Prompt / Terminal

- Used to execute the Python script and interact with the chatbot during testing.

Theory

A **Rule-Based AI Chatbot** is a simple artificial intelligence system that responds to users based on predefined rules and conditions. It does not learn from data; instead, it follows instructions written by the programmer.

The chatbot receives input from the user, checks the input using **if-else statements**, and provides a suitable response. A **while loop** is used to keep the chatbot running continuously until the user enters an exit command.

Key Concepts Used:

- **Artificial Intelligence (AI):** Enables machines to simulate basic human interaction.
- **Conditional Statements (if-else):** Used to make decisions based on user input.
- **Loops:** Allow the chatbot to continue the conversation.
- **String Handling:** Helps process and compare user-entered text.
- **Variables:** Variables are used to store data temporarily in memory.
- **Input:** Input is the data entered by the user into the chatbot.
- **Output:** Output is the response generated and displayed by the chatbot.

Algorithm

Step 1: Start the Program

- Execute the Python program.
- The chatbot becomes ready to interact with the user.

Step 2: Display Welcome Message

- Show a greeting message such as:
"Welcome to Rule-Based AI Chatbot"
- Inform the user how to exit the chatbot.

Step 3: Enter Continuous Loop

- Use while True to keep the chatbot running continuously.
- This allows the chatbot to respond to multiple user queries.

Step 4: Take User Input

- Ask the user to enter a message.
- Store the input in a variable.

Step 5: Convert Input to Lowercase

- Use .lower() to convert the user's input to lowercase.
- This ensures inputs like "Hello", "HELLO", and "hello" are treated the same.

Step 6: Check for Greeting

- If the user enters greetings such as "hi", "hello", or "hey", the chatbot responds with a greeting message.

Step 7: Check Other Predefined Questions

- Compare the user's input with predefined questions.
- Provide the corresponding response if a match is found.

Step 8: Check for Exit Command

- If the user enters "bye", "exit", or "quit", display a goodbye message and stop the chatbot.

Step 9: Handle Unknown Inputs

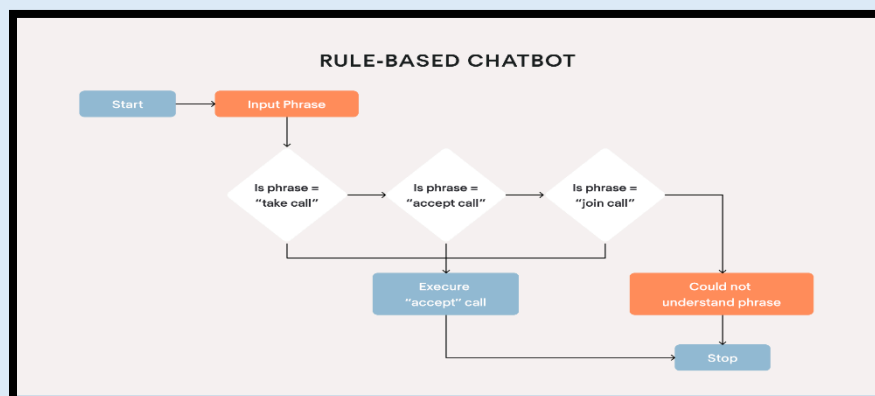
- If the input does not match any predefined rule, display a default message such as:
"Sorry, I don't understand that."

Step 10: Repeat the Process

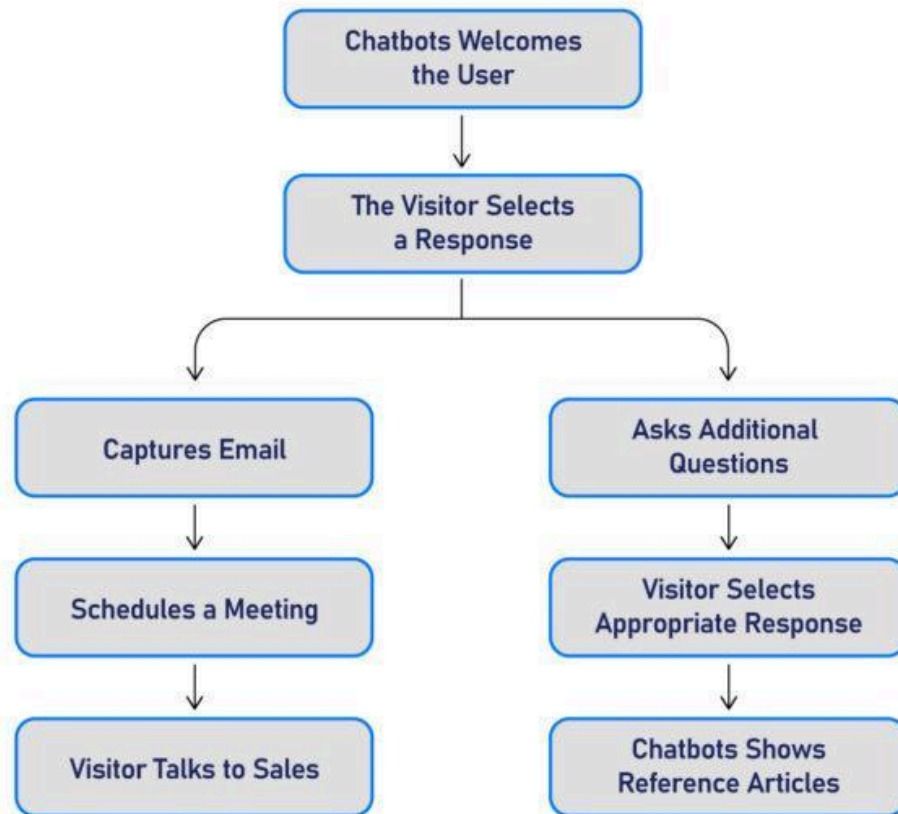
- Return to Step 4 and wait for the next user input.
- Continue until the user enters an exit command.

Step 11: End the Program

- When an exit command is received, terminate the loop using break.
- The chatbot stops running.



Flowchart



Code with Python

```
main.py  [ ] [ ] [ ] Share Run

1 print("👋 Welcome to Rule-Based AI Chatbot")
2 print("Type 'bye' to exit.\n")
3 while True:
4     user = input("You: ").lower()
5     if user in ["hi", "hello", "hey"]:
6         print("Bot: Hello! How can I help you?")
7     elif user == "how are you":
8         print("Bot: I am doing great!")
9     elif user == "what is your name":
10        print("Bot: My name is RuleBot.")
11    elif user == "help":
12        print("Bot: You can ask me simple questions.")
13    elif user in ["bye", "exit", "quit"]:
14        print("Bot: Goodbye! Have a nice day.")
15        break
16    else:
17        print("Bot: Sorry, I don't understand that.")
```

Output

```
Output

👋 Welcome to Rule-Based AI Chatbot
Type 'bye' to exit.

You: hello
Bot: Hello! How can I help you?
You: what is your name
Bot: My name is RuleBot.
You: bye
Bot: Goodbye! Have a nice day.

=== Code Execution Successful ===
```

Advantages of Rule based AI Chatbot

1. Easy to Develop

The chatbot is simple to create because it uses basic Python concepts like loops and conditional statements.

2. Fast Response

It provides instant replies since responses are predefined.

3. Beginner Friendly

It helps beginners understand the fundamentals of Artificial Intelligence and chatbot development.

4. No Internet Required

The chatbot works offline because all responses are stored within the program.

5. Low Resource Usage

It does not require powerful hardware or complex AI models.

Limitations of Rule Based AI Chatbot

1. Limited Responses

The chatbot can only answer questions that are already programmed.

2. No Learning Ability

It cannot learn from user interactions or improve its responses automatically.

3. Poor Understanding

It cannot understand complex sentences or different ways of asking the same question.

4. No Context Awareness

The chatbot cannot remember previous conversations.

5. Not Suitable for Advanced Tasks

It cannot perform intelligent reasoning like modern AI chatbots.

Future Enhancements Rule Based AI Chatbot

1. Add More Commands

Increase the number of questions and responses to make the chatbot more useful.

2. Use Dictionaries

Replace multiple if-else statements with dictionaries for better code management.

3. Add GUI Interface

Create a graphical user interface using Tkinter for a better user experience.

4. Integrate Speech Recognition

Allow users to interact with the chatbot using voice commands.

5. Implement NLP

Use Natural Language Processing (NLP) to understand user inputs more accurately.

6. Add Machine Learning

Enable the chatbot to learn from conversations and improve over time.

7. Store Conversation History

Save previous chats to provide a more personalized experience.

8. Connect to APIs

Integrate weather, news, calculator, or other APIs to provide real-time information.

Result

A rule-based AI chatbot was successfully developed using Python. The chatbot responds to greetings, answers predefined questions, handles exit commands, and operates continuously using control flow and decision-making logic. This satisfies the project requirements.

Conclusion

The Rule-Based AI Chatbot demonstrates the fundamental concepts of Artificial Intelligence, including control flow, decision-making, and user interaction. It serves as a foundation for building more advanced AI systems in the future.

THANK YOU